AC and DC Bridges

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Bridges

- There are two types of bridges:

  DC Bridge: Is used to measure the resistance. Battery is used as a source of energy and galvanometer is used as a detector.

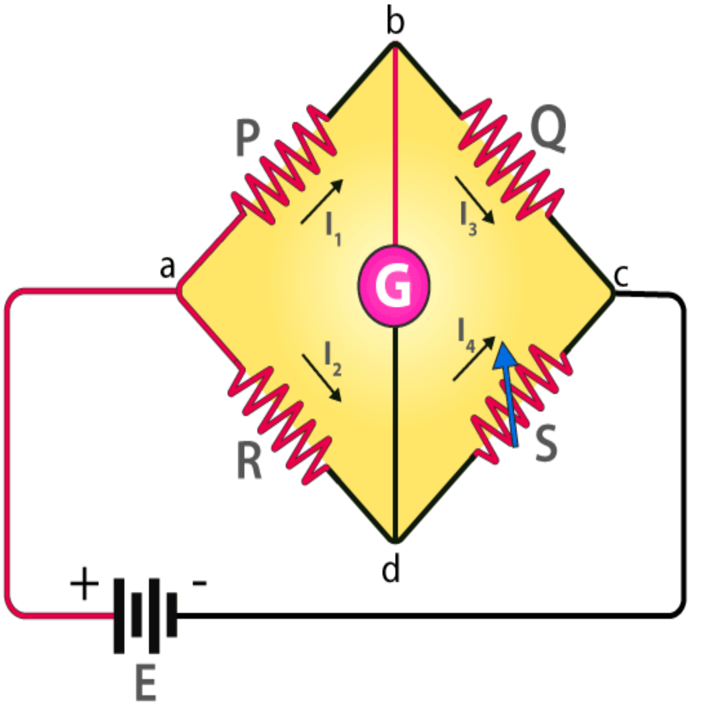
  AC Bridge: Is used to measure inductance, capacitance, loss factor etc. AC source (Supply or Oscillator) is used as source of energy and head phone is used as detector.

DC Bridge

- Wheatstone and Kelvin double bridge are the examples of DC bridge.
- Wheatstone bridge is used to measure medium range of resistance (1 Ω to 0.1 MΩ). Other methods which are used to measure medium resistance are **Ammeter-voltmeter, substitution, ohmmeter** method.
- Kelvin double bridge is used to measure low range of resistance (range of 1 Ω). Other methods which are used to measure low resistance are **Ammeter-voltmeter and potentiometer** methods.

Wheatstone bridge

- A Wheatstone bridge circuit consists of four arms.
- Two arms consist of known resistances while the other two arms consist of an unknown resistance and a variable resistance.
- The circuit also consists of a galvanometer and a Battery.
- The battery is attached between point ‘*a*’ and ‘c’ while galvanometer is connected between the points ‘b’ and ‘*d*’.
- The current that flows through the galvanometer depends on the potential difference between point ‘b’ and ‘d’.



- In the figure, R is the unknown resistance, and the S is called the ‘standard arm’ of the bridge and the P and Q are called the ‘ratio arms’ of the bridge.
- The current divides into two equal magnitude as I_1 and I_2.
- The following condition exists when the current through a galvanometer is zero,

$$\mathbf{I_1 P = I_2 R} \qquad \mathbf{(1)}$$

- The currents in the bridge, in a balanced condition, is expressed as follows:

$$\mathbf{I_1 = I_3 = E/\,(P+Q)} \qquad \mathbf{I_2 = I_4 = E/\,(R+S)}$$

Here, *E* is the emf of the battery.

- By substituting the value of I_1 and I_2 in equation (1), we get

$$\mathbf{P[E/(P+Q)] = R[E/(R+S)]}$$

$$\mathbf{P/(P+Q) = R/(R+S)}$$

$$\mathbf{P(R+S) = R(P+Q)}$$

$$\mathbf{PR+PS = RP+RQ}$$

$$\mathbf{PS = RQ} \qquad \mathbf{(2)}$$

$$\mathbf{R = (P/Q)\times S} \qquad \mathbf{(3)}$$

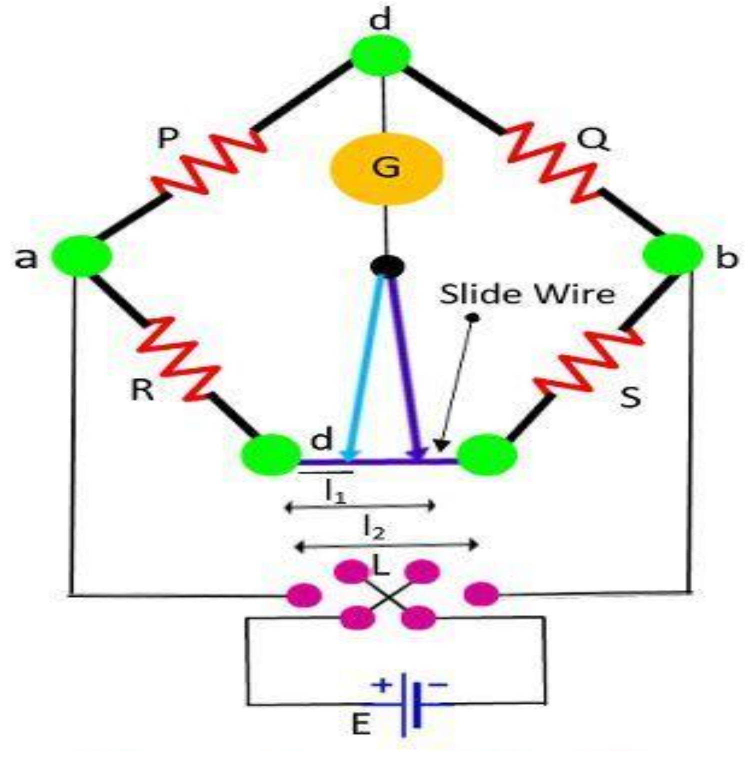
- Equation (2) shows the balanced condition of the bridge. Equation (3) determines the value of the unknown resistance.

Factors consider for accurate measurement

- **Resistance of connecting leads**-A lead of 22 SWG wire having a length of 25 cm has a resistance of about 0.012 Ω.
- **Thermo-electric effects-**Thermoelectric emfs are present in the measuring circuit. It must be considered because its effect on the galvanometer is same as unbalance. Its effect may be eliminated by reversing the battery connections to the bridge and adjusting the balance until no change in galvanometer deflection can be observed on the reversal.
- **Temperature Effects-** The serious errors is produced by change of resistance due to change of temperature especially if resistance temperature coefficient is high.
- **Contact Resistance-** Serious errors may occur due to contact resistance between switches or wires.

Carey-Foster Slide-Wire Bridge

- This bridge is used for determining the difference between the standard and the unknown resistance or comparison of two nearly equal resistance.
- The P, Q, R and S are four resistors used in the bridge.
- Resistances 'P' and 'Q' are known while the 'R' is unknown resistor and 'S' is the standard resistor.
- The slide wire of length L is placed between the resistance R and S.
- The resistors 'P' and 'Q' are adjusted so that the ratio of P/Q is approximately equal to the ratio of R/S.
- Exact balance is obtained by adjustment of the sliding contact on the slide-wire.
- Let l_1 is the distance from the left at which the balanced is obtained.
- The 'R' and 'S' are interchanged and this time the balanced is obtained by sliding the contact at the distance of l_2.



- Equation for first balance-

$$\frac{P}{Q} = \frac{R + l_1 r}{S + (L - l_1) r} \qquad (1)$$

The ‘r’ is the resistance per unit length of the sliding wire.

- After interchanging the ‘R’ and ‘S’, the balance equation of the bridge is

$$\frac{P}{Q} = \frac{S + l_2 r}{R + (L - l_2)} \qquad (2)$$

Now, for first balance,

$$\frac{P}{Q} + 1 = \frac{R + l_1 r + S + (L - l_1) r}{S + (L - l_1) r} = \frac{R + S + Lr}{S + (L - l_1) r} \qquad (3)$$

also, for second balance,

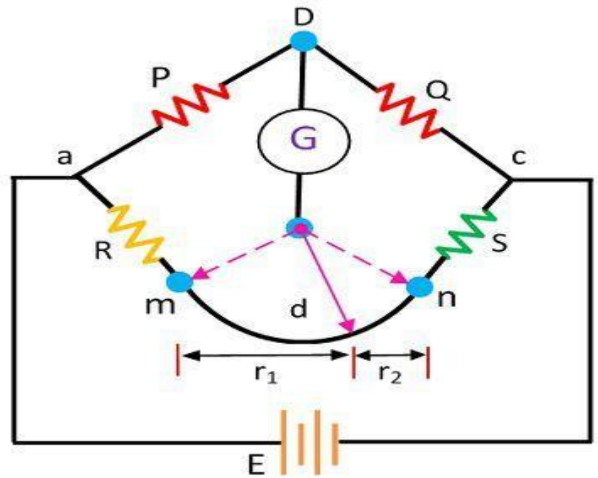
$$\frac{P}{Q} + 1 = \frac{S + l_2 r + R + (L - l_2) r}{R + (L - l_2) r} = \frac{S + R + Lr}{R + (L - l_2) r} \qquad (4)$$

From eq. (3) and (4)

$S + (L - l_1) r = R + (L - l_2) r$ Hence $R - S = (l_2 - l_1) r$

Kelvin bridge

- A Kelvin bridge is a modified version of the Wheatstone bridge, which can measure resistance values in the range between 1 to10^{-6} ohms with high accuracy.



- The P, Q, R and S are four resistors used in the bridge. Resistors ‘P’ and ‘Q’ are known valued while the ‘R’ is unknown and ‘S’ is standard.
- A lead of resistance ‘r’ is connected between unknown resistor ‘R’ and standard resistor ‘S’.
- Two galvanometer connections at point ‘m’ and ‘n’ may be made.
- When the galvanometer is connected to point ‘m’, resistance ‘r’ of lead is added to the standard resistance ‘S’ and obtain high value of unknown resistance.

- When the galvanometer is connected to point ‘n’, resistance ‘r’ of lead is added to the unknown resistance ‘R’ and obtain low value of unknown resistance.
- If galvanometer is connected at point ‘d’ (as shown by full line), the resistance ‘r’ is divided into two parts, ‘r_1’ and ‘r_2’ such that-

$$\frac{r_1}{r_2} = \frac{P}{Q}. \qquad (5)$$

- If bridge is under balanced condition then

$$R + r_1 = \frac{P}{Q}.(S + r_2) \qquad (6)$$

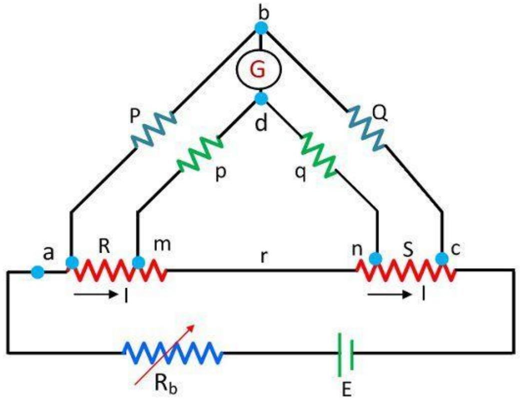
- After simplification of eq. (5) and (6)

$$R = \frac{P}{Q}.S$$

- The above equation shows that if the galvanometer connects at point d then the resistance of lead will not affect their results.

Kelvin double bridge

- For obtaining the desired result, the actual resistance of exact ratio connects between the point ‘m’ and ‘n’ and the galvanometer connects at the junction of the resistor.



- The ‘P’ and ‘Q’ is the first ratio arm and ‘p’ and ‘q’ is the second ratio arm.
- The second set of ratio arm, ‘p’ and ‘q’ are used to connect the galvanometer to a point ‘d’ at the appropriate point ‘m’ and ‘n’ to eliminate the effect of connecting lead of resistance ‘r’ between the unknown resistance ‘R’ and the standard resistance ‘S’.
- The ratio of ‘p/q’ is made equal to the ratio of ‘P/Q’.
- Under balance condition, zero current flows through the galvanometer. The potential difference (E_{ab}) between the point ‘a’ and ‘b’ is equivalent to the voltage drop between the points E_{amd}.

$$E_{ab} = \frac{P}{P+Q} E_{ac}$$

$$E_{ac} = I\left[R + S + \frac{(p+q)r}{p+q+r}\right] \cdots$$

$$E_{amd} = I\left[R + \frac{p}{(p+q)}\left\{\frac{(p+q)r}{p+q+r}\right\}\right] = I\left[R + \frac{pr}{p+q+r}\right]$$

- For zero galvanometer deflection

$$E_{ab} = E_{amd}$$

$$\frac{P}{P+Q} \cdot I\left[R + S + \frac{(p+q)r}{p+q+r}\right] = I\left[R + \frac{pr}{p+q+r}\right]$$

$$R = \frac{P}{Q}.S + \frac{pr}{p+q+r}\left[\frac{P}{Q} - \frac{p}{q}\right]$$

As we known, P/Q = p/q then above equation becomes $R = \frac{P}{Q} \cdot S$

The equation shows that the result obtains from the Kelvin double bridge is free from the impact of the connecting lead resistance.

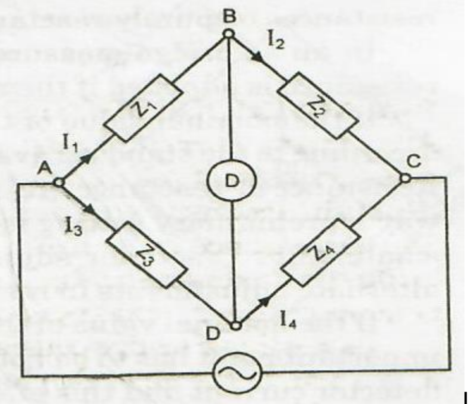
- A resistance of approximately 3000 Ω is needed to balance a bridge. It is obtained on a 5 dial resistance box having steps of 1000, 100, 10, 1 and 0.1 Ω. The measurement is to be guaranteed to 0.1%. For this accuracy, how many of these dials would it be worth adjusting?
- Each of the ratio arms of a laboratory type Wheatstone bridge has guaranteed accuracy of $\pm$0.05%, while the standard arm has a guaranteed accuracy of $\pm$0.01%. The ratio arms are set at 1000 Ω and the bridge is balanced with standard arm adjusted to 3154 Ω. Determine the upper and lower limits of the unknown resistance, based upon the guaranteed accuracies of the known bridge arms.
- In a carey-Foster's bridge a resistance of 1.0125 Ω is compared with a standard resistance of 1.0000 Ω, the slide wire has a resistance of 0.250 Ω in 100 divisions. The ratio arms nominally each 10Ω, are actually 10.05 and 9.95 Ω respectively. How far (in scale division) are the balance positions from those which would obtain of ratio arms were true to their nominal value? The slide wire is 100 cm long.
- A Kelvin double bridge each of the ratio arms P = Q = p = q =1000 Ω. The emf of the battery is 100 V and a resistance of 5 Ω is included in the battery circuit. The galvanometer has a resistance of 500 Ω and the resistance of the link connecting the unknown resistance to the standard resistance may be neglected. The bridge is balanced when the standard resistance S = 0.001 Ω. (a) Determine the value of unknown resistance. (b) Determine the current through the unknown resistance R at balance. (c) Determine the deflection of the galvanometer when the unknown resistance R is changed by 0.1 % from its value at balance. The galvanometer has a sensitivity of 200 mm/μA.

AC Bridges

- AC bridges are used to measure inductance, capacitance, storage factor, loss factor etc.
- **Sources-**
  - For low frequency applications, power line may act as the source of supply.
  - For higher frequencies electronic oscillators are used as the source of supply.
- **Detectors-**
  - **Head Phones**: are widely used as detectors frequencies from 250 Hz- 4 kHz at low power level.
  - **Vibration Galvanometer**: is extremely used for power and low audio frequency ranges. Vibration Galvanometer is used in the range of 5 Hz to 1000 Hz but are most commonly used below 200 Hz as below this frequency they are most sensitivity than the head phones.
  - **Tuneable amplifier detectors:** are the most versatile of the detectors. This detector can be used over a frequency range of 10 Hz to 100 KHz.

General Equation for Bridge Balance

- The bridge consists of Z_1, Z_2, Z_3 and Z_4.



- Under balanced condition, no current should flow through the detector.
- This requires that the potential difference between points B and D should be zero.

$E_1 = E_3$ (1)

$I_1 Z_1 = I_3 Z_3$ (2)

$I_1 = I_2 = E / (Z_1 + Z_2)$ (3)

$I_3 = I_4 = E / (Z_3 + Z_4)$ (4)

After simplification-

$Z_1 Z_4 = Z_2 Z_3$ (5) (Series elements)

In case of parallel arrangement, admittances are taken instead of impedance

$Y_1 Y_4 = Y_2 Y_3$ (6) (Parallel elements)

- Impedance can be written as $Z = Z\angle\theta$ in polar form
  - Where Z represents the magnitude and θ represent the phase angle of the complex impedance.
- Eq. (5) can be written as

$$(Z_1\angle\theta_1)(Z_4\angle\theta_4) = (Z_2\angle\theta_2)(Z_3\angle\theta_3) \qquad (7)$$

- Thus for balanced condition-

$$Z_1\, Z_4\, \angle(\theta_1 + \theta_4) = Z_2\, Z_3\, \angle(\theta_2 + \theta_3) \qquad (8)$$

- Eq. (8) shows that two conditions must be satisfied simultaneously when balancing an **AC bridge**. The first condition is that the magnitude of impedances satisfies the relationship :

$$Z_1\, Z_4 = Z_2\, Z_3 \qquad (9)$$

The condition in above equation is called **magnitude criteria.**

- The second condition is that the phase angles of impedances satisfy the relationship :

$$\angle(\theta_1 + \theta_4) = \angle(\theta_2 + \theta_3) \qquad (10)$$

This condition is known as **phase criteria**.

- The phase angles are positive for an inductive impedance and negative for capacitive impedance.

- In terms of rectangular coordinates, impedance can be written as

$$Z_1 = R_1 + jX_1 \qquad\qquad Z_2 = R_2 + jX_2$$

$$Z_3 = R_3 + jX_3 \qquad \text{and} \qquad Z_4 = R_4 + jX_4 \qquad (11)$$

- For balance, $$Z_1 Z_4 = Z_2 Z_3 \qquad (12)$$

or $$(R_1 + jX_1)(R_4 + jX_4) = (R_2 + jX_2)(R_3 + jX_3) \qquad (13)$$

or $$R_1 R_4 - X_1 X_4 + j(X_1 R_4 + X_4 R_1) = R_2 R_3 - X_2 X_3 + j(X_2 R_3 + X_3 R_2)$$

- The above equation is a complex equation and a complex equation is satisfied only if real and imaginary parts of each side of the equation are separately equal. Thus, for balance,

$$R_1 R_4 - X_1 X_4 = R_2 R_3 - X_2 X_3 \qquad (14)$$

$$X_1 R_4 + X_4 R_1 = X_2 R_3 + X_3 R_2 \qquad (15)$$

- Thus there are two independent conditions for balance and both of them must be satisfied for the AC bridge to be balanced.

- The four impedance of an AC bridge are

  $Z_1 = 400\ \Omega\angle 50^o$; $\quad$ $Z_2 = 200\ \Omega\ \angle 40^o$

  $Z_3 = 800\ \Omega\ \angle -50^o$ $\quad$ $Z_4 = 400\ \Omega\ \angle 20^o$.

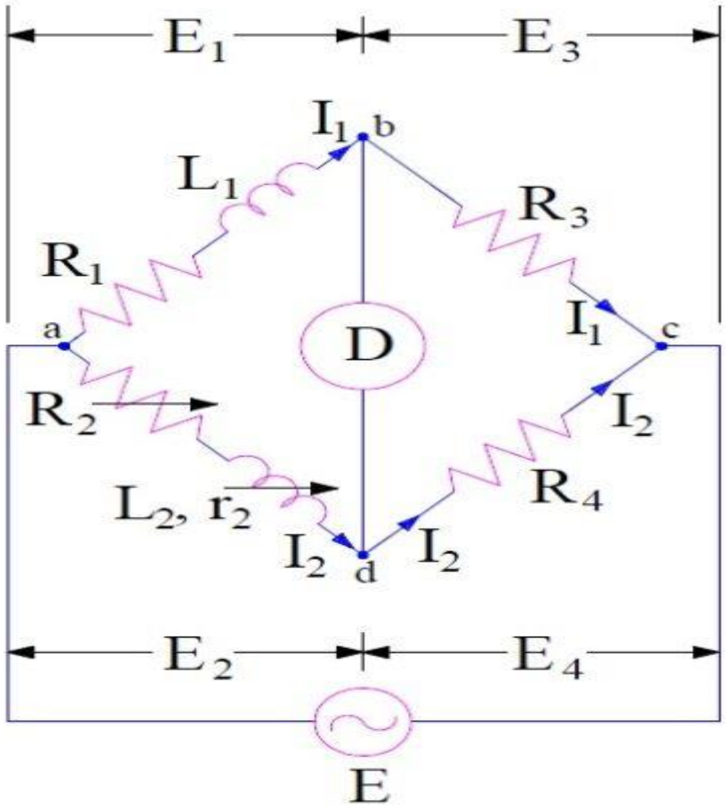
  Find out whether the bridge is balance under these conditions or not.
- An AC bridge circuit is working at 1000 Hz. Arm AB is a 0.2μF pure capacitance; arm BC is a 500 Ω pure resistance; arm CD contains an unknown impedance and arm DA has a 300 Ω resistance in parallel with a 0.1 μF capacitor. Find the R and C or L constant of arm CD considering it as a series circuit.

AC Bridges

- **Maxwell Inductance Bridge**
- **Maxwell Inductance-Capacitance Bridge**
- **Hay’s Bridge**
- Anderson’s Bridge
- Own’s Bridge
- **De sauty’s Bridge**
- **Schering Bridge**
- Heaviside Mutual Inductance Bridge
- Campbell’s Mutual Inductance Bridge

Maxwell Inductance Bridge

- Maxwell Bridge is used for the measurement of Self-Inductance.
- In this method, unknown inductance is determined by comparing it with a standard self inductance.



L_1 – unknown inductance of resistance R_1.
L_2 – Variable inductance of fixed resistance r_1.
R_2 – variable resistance connected in series with inductor L_2.
R_3, R_4 – known non-inductance resistance

- Under balanced condition, current flowing through detector D is zero.
- For a balanced bridge the multiplication of impedances of opposite arms must be equal.

  Impedance of arm ab, $Z_1 = (R_1 + j\omega L_1)$

  Impedance of arm cd, $Z_4 = R_4$

  Impedance of arm ad, $Z_2 = (R_2 + r_2 + j\ \omega\ L_2)$

  Impedance of arm bc, $Z_3 = R_3$
- Hence for balanced bridge,

$$Z_1 Z_4 = Z_2 Z_3$$

$$(R_1 + j\ \omega\ L_1) \times R_4 = (R_2 + r_2 + j\ \omega\ L_2) \times R_3$$

$$R_1 R_4 - R_2 R_3 - r_2 R_3 + j\ \omega (L_1 R_4 - L_2 R_3) = 0$$

- Equating real and imaginary part we get,

$$R_1 R_4 - R_2 R_3 - r_2 R_3 = 0 \quad \ldots\ldots\ldots\ldots\ldots\ldots(16)$$

$$\text{and} \quad (L_1 R_4 - L_2 R_3) = 0 \quad \ldots\ldots\ldots\ldots\ldots\ldots(17)$$

- From (16)

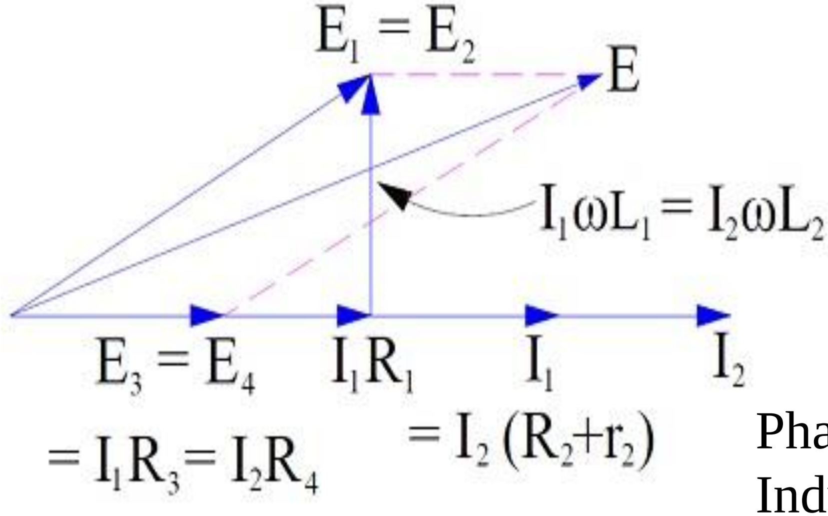
$$R_1R_4 = R_2R_3+r_2R_3$$
$$= R_3(R_2+r_2)$$

Hence, $\mathbf{R_1 = (R_3/R_4)(R_2+r_2)}$

- Now from (17),

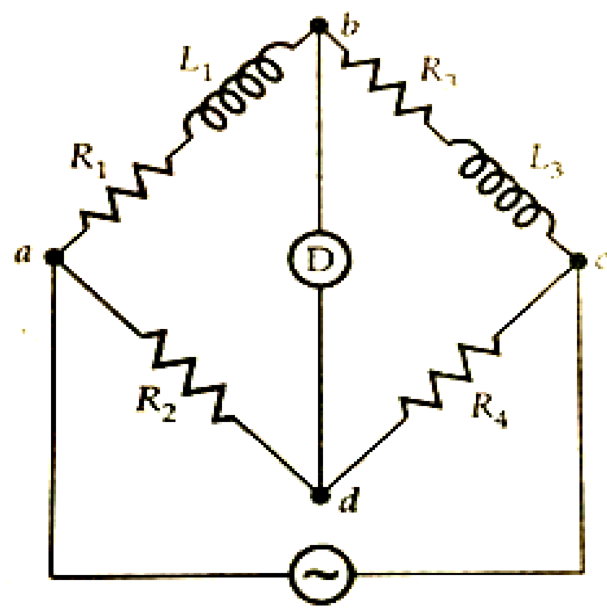
$$L_1R_4 = L_2R_3$$

- Hence, $\mathbf{L_1 = L_2R_3 / R_4}$
- Thus unknown inductance L_1 and its resistance R_1 may be calculated.



Phasor Diagram of Maxwell Inductance Bridge

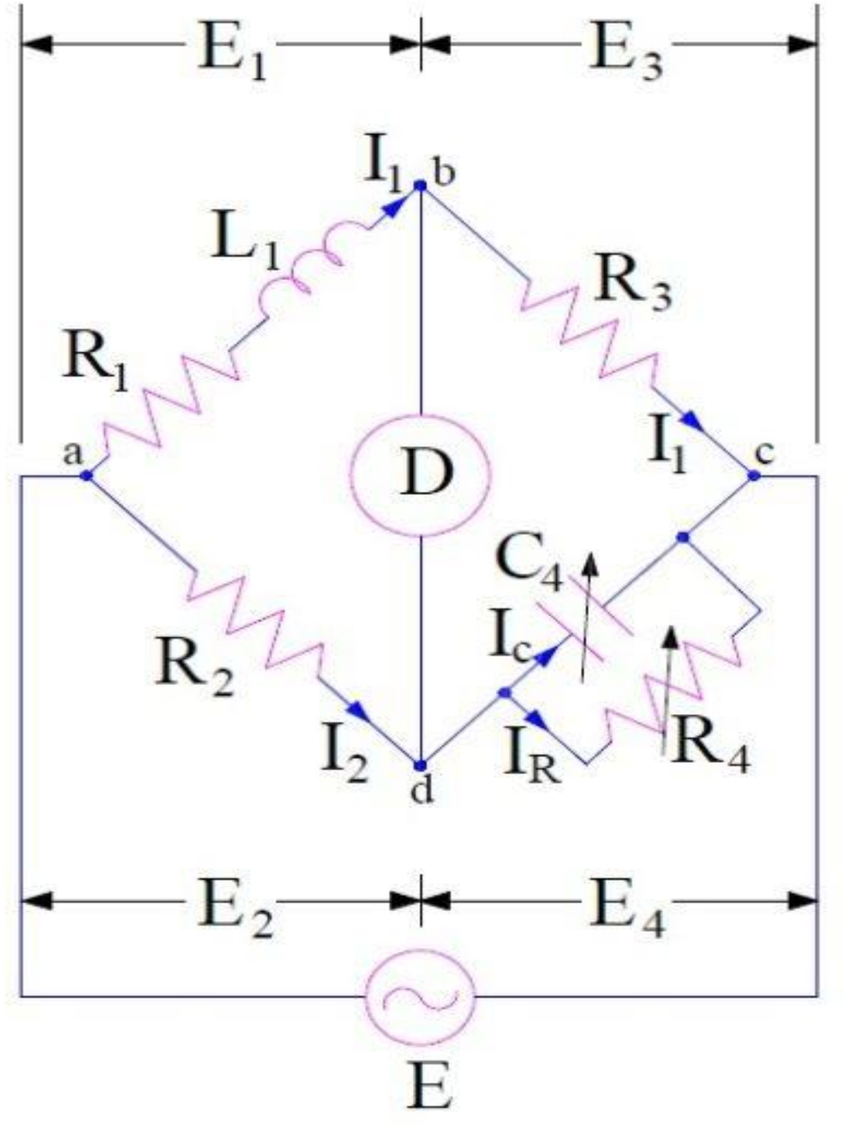
- An inductance of 0.22 H and 20 Ω resistance is measured by comparison with a fixed standard inductance of 0.1 H and 40 Ω resistance. They are connected as shown in figure. The unknown inductance is in arm ‘ab’ and the standard inductance is arm ‘bc’, a resistance of 750 Ω is connected in arm ‘cd’ and a resistance whose amount is not known is in arm Find the resistance of arm ‘da’ and show any necessary and practical additions required to achieve both resistive and inductive balance



Maxwell Inductance-Capacitance Bridge or Maxwell Wien Bridge

- In this method, the unknown inductance is measured by comparison with standard known capacitance.

- L_1 = Unknown inductance with resistance R_1
- C_4 = Variable standard capacitor
- R_4 = Variable known resistance
- R_2 and R_3 = Known fixed resistance



- Impedance of arm ab, $Z_1 = (R_1 + j\omega L_1)$
- Impedance of arm ad, $Z_2 = R_2$
- Impedance of arm bc, $Z_3 = R_3$
- Impedance of arm cd, $Z_4 = R_4 / (1 + j\,\omega\, C_4 R_4)$

- For bridge to be balance,

$$Z_1 Z_4 = Z_2 Z_3$$

$$(R_1 + j\,\omega\, L_1) \times [R_4 / (1 + j\,\omega\, C_4 R_4)] = R_2 R_3$$

$$R_1 R_4 - R_2 R_3 + j\,\omega\,(L_1 R_4 - R_2 R_3 C_4 R_4) = 0$$

- Equating real and imaginary parts-

$$\mathbf{R_1 = R_2 R_3 / R_4}$$

$$\textbf{and } \mathbf{L_1 = R_2 R_3 C_4}$$

- The quality factor of inductor may also be calculated as

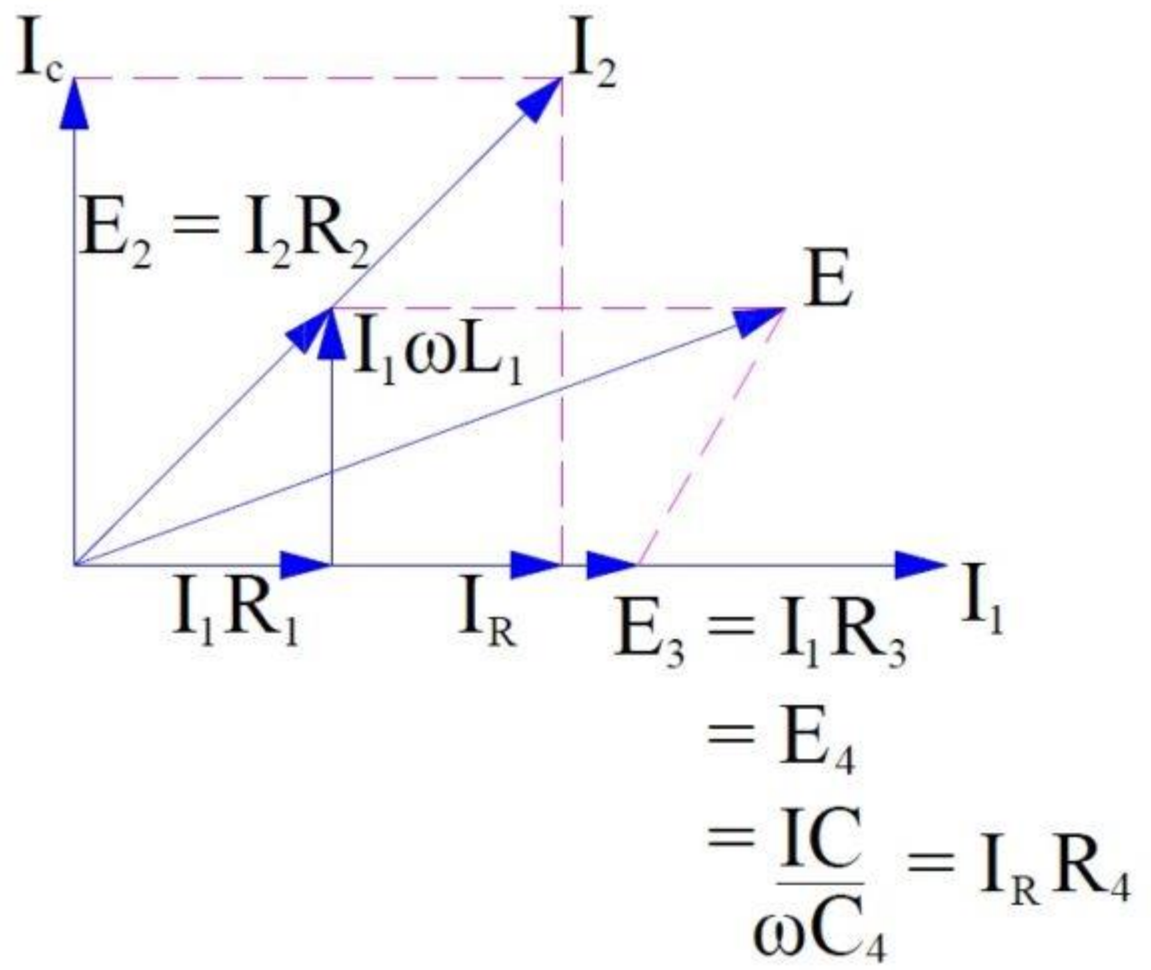
$$Q = \omega\, L_1 / R_1$$

$$= \omega\, R_2 R_3 C_4 / R_1$$

- Since $R_4 = R_2 R_3 / R_1$, hence

$$Q = \omega\, C_4 R_4$$

- The Phasor diagram of Maxwell Inductance Capacitance Bridge is shown below.

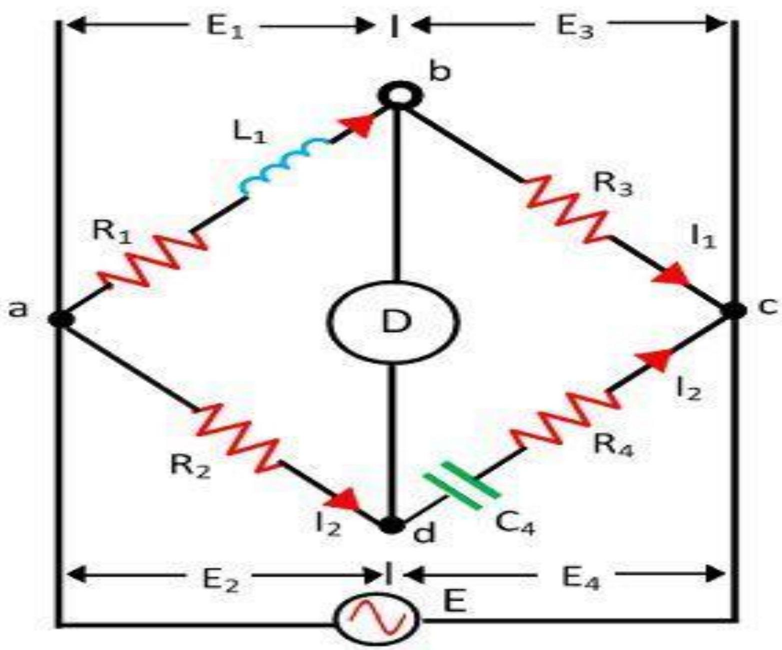


- **Advantage:**
  - The expression of inductance is independent of frequency.
  - A wide range of inductance can be measured at power and audio frequencies.
  - The expression for inductance is simple and can easily be calculated.
- **Disadvantage:**
  - Since Maxwell Inductance-Capacitance Bridge uses variable standard capacitor, it is very expensive to get variable standard capacitor.
- **Applicability of Maxwell Bridge:**
  - Maxwell Bridge is suitable for the measurement of inductance with medium value of Q ($1<Q<10$). This method is not suitable for measurement of inductance with high value of quality factor Q. Since $Q = \omega C_4 R_4$, higher value of resistance R_4 and capacitance C_4 for measurement of high Q coil which are very expensive.

- In case of Maxwell Inductance-Capacitance Bridge $R_2 = 600\ \Omega$, $R_3 = 400\ \Omega$, $R_4 = 1000\ \Omega$ and $C_4 = 0.5\ \mu F$. Determine the value of unknown R and L of the inductor and Q-factor of the coil.
- In case of Maxwell Inductance-Capacitance Bridge, if resistors R_2, R_3 and R_4 can have a variation of $\pm$ 0.2 % and C have variation of $\pm$ 0.1% from their nominal values, estimate the percentage error in the evaluation of R and L.

Hay's Bridge

- The **Hay's bridge** is used for determining the self-inductance of the circuit.
- This bridge is the advanced form of Maxwell's bridge which is used to measure medium quality of Q-factor.
- Hay's bridge is used for measuring high quality of Q-factor.
- In Hay's bridge a capacitor is connected in series with the resistance.



- The unknown inductor L_1 is placed with the resistance R_1 in the arm '**ab**'.
- This unknown inductor is compared with the standard capacitor C_4 connected across the arm '**cd**'.
- The resistance R_4 is connected in series with the capacitor C_4.
- Two non-inductive resistors R_2 and R_3 are connected in the arm '**ad**' and '**bc**' respectively.
- The C_4 and R_4 are adjusted for making the bridge in the balanced condition.
- When the bridge is under balanced condition, no current flows through the detector.
- At balanced condition-

$$Z_1 Z_4 = Z_2 Z_3 \qquad (1)$$

Impedance of arm '**ab**' $Z_1 = R_1 + j\omega L_1$

Impedance of arm '**ad**' $Z_2 = R_2$

Impedance of arm '**bc**' $Z_3 = R_3$

Impedance of arm '**cd**' $Z_4 = R_4 + 1/j\omega C_4 \qquad (2)$

$$(R_1 + j\omega L_1)(R_4 - j/\omega C_4) = R_2 R_3 \quad (3)$$

$$R_1 R_4 + \frac{L_1}{C_4} + j\omega L_1 R_4 - \frac{jR_1}{\omega C_4} = R_2 R_3 \quad (4)$$

- Separating the real and imaginary term-

$$R_1 R_4 + \frac{L_1}{C_4} = R_2 R_3 \quad (5)$$

$$L_1 = \frac{R_1}{\omega^2 R_4 C_4} \quad (6)$$

- Put the value of eq. (6) into eq. (5)

$$L_1 = \frac{R_2 R_3 C_4}{1 + \omega^2 R_4^2 C_4^2} \qquad R_1 = \frac{\omega^2 C_4^2 R_2 R_3 R_4}{1 + \omega^2 R_4^2 C_4^2} \quad (7)$$

- The quality factor of the coil is

$$Q = \frac{\omega L_1}{R_1} = \frac{1}{\omega R_4 C_4} \quad (8)$$

- The equation of the unknown inductance and resistance contain frequency term. Thus for finding the value of unknown inductance the frequency of the supply must be known accurately.
- For the high-quality factor, the frequency does not play an important role.

$$Q = \frac{\omega L_1}{R_1} = \frac{1}{\omega R_4 C_4} \qquad (9)$$

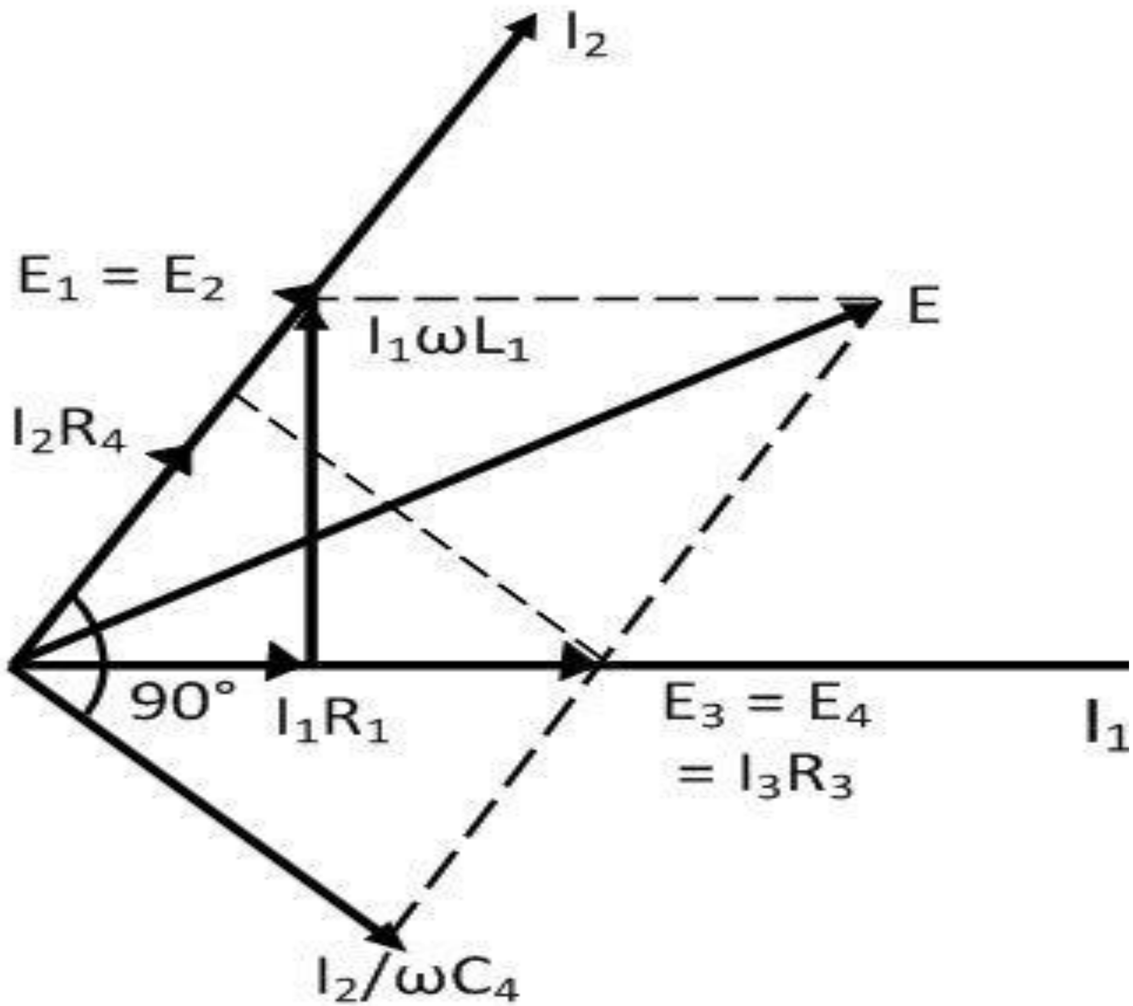
- Substituting the value of Q in the equation (7) of unknown inductance, we get

$$L_1 = \frac{R_2 R_3 C_4}{1 + (1/Q)^2} \qquad (10)$$

- For greater value of Q the 1/Q is neglected and hence the equation become

$$L_1 = R_2 R_3 C_4 \qquad (11)$$

- Phasor Diagram of Hay’s Bridge



- The current passes through the arm ‘**ab**’ produces a potential drop $\mathbf{I_1R_1}$ which is also in the same phase of $\mathbf{I_1}$. The total voltage drop across the arm ‘**ab**’ is determined by adding the voltage $\mathbf{I_1R_1}$ and $\boldsymbol{\omega}\mathbf{L_1I_1}$.
- The voltage drops across the arm ‘**ab**’ and ‘**ad**’ are equal. The voltage drop $\mathbf{E_1}$ and $\mathbf{E_2}$ are equal in magnitude and phase and hence overlap each other. The current $\mathbf{I_2}$ and $\mathbf{E_2}$ are in the same phase as shown in the figure above.
- The current $\mathbf{I_2}$ flows through the arms ‘**cd**’ and produces the $\mathbf{I_2R_4}$ voltage drops across the resistance and $\mathbf{I_2}/\boldsymbol{\omega}\mathbf{C_4}$ voltage drops across the capacitor $\mathbf{C_4}$. The capacitance voltage across $\mathbf{C_4}$ lags by the currents 90^o.
- The voltage drops across the resistance $\mathbf{C_4}$ and $\mathbf{R_4}$ give the total voltage drops across the arm ‘**cd**’. The sum of the voltage “$\mathbf{E_1}$ and $\mathbf{E_3}$” or “$\mathbf{E_2}$ and $\mathbf{E_4}$” gives the voltage drops **E**.

- **Advantages**
- The Hays bridges give a simple expression for the unknown inductances and are suitable for the coil having the quality factor greater than the 10 ohms.
- The Hay's bridge uses small value resistance and capacitance for determining the Q factor.
- **Disadvantages**
- The only disadvantage of this type of bridge is that it is not suitable for the measurement of the coil having the quality factor less than 10.
- Frequency of supply should be known accurately to find out the value of resistance, inductance and Q-factor.

De Sauty's Bridge

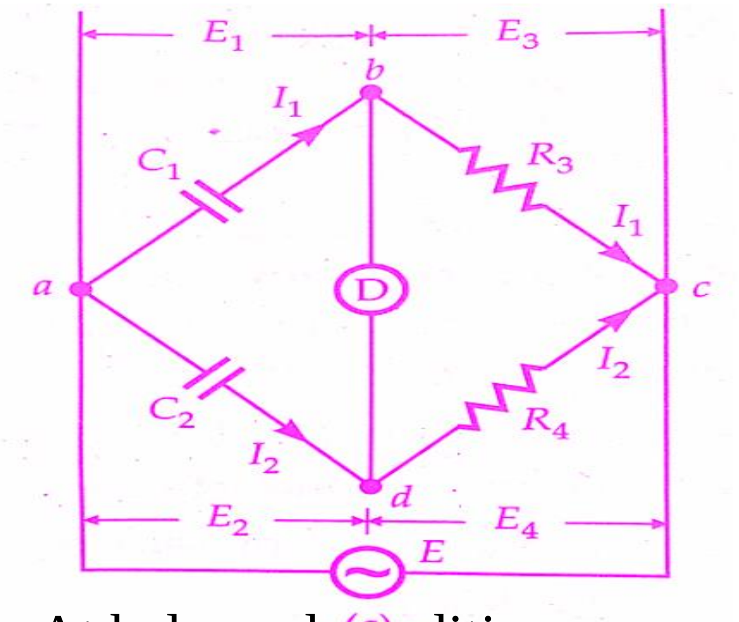
- De Sauty's Bridge is the simplest method to find out the value of unknown capacitance.

- The value of unknown capacitance is obtained by comparing with standard capacitances.

C_1 = Capacitor whose capacitance is to be measured

C_2 = A standard capacitor, and

R_3 and R_4 = non-inductive resistors.



- At balanced condition-

$Z_1 Z_4 = Z_2 Z_3$ (1)

Impedance of arm '**ab**' $Z_1 = 1/j\omega C_1$

Impedance of arm '**ad**' $Z_2 = 1/j\omega C_2$

Impedance of arm '**bc**' $Z_3 = R_3$

Impedance of arm '**cd**' $Z_4 = R_4$

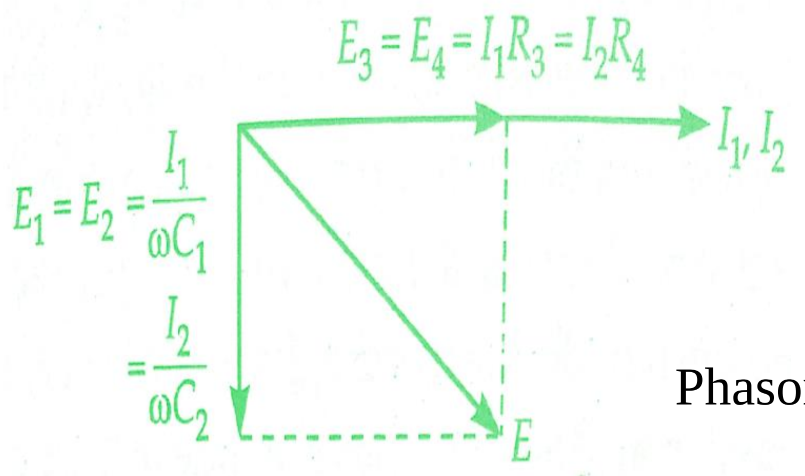
(2)

- Put the expression of eq. (2) in to eq. (1)

$$\left(\frac{1}{j\omega C_1}\right) R_4 = \left(\frac{1}{j\omega C_2}\right) R_3 \qquad (3)$$

$$C_1 = C_2\, R_4 / R_3 \qquad (4)$$

- The balance can be obtained by varying either R_3 or R_4.
- It is impossible to obtain balance if both the capacitors are not free from dielectric loss.
- Thus with **DeSauty's Bridge,** only loss-less capacitors like air capacitors can be compared.

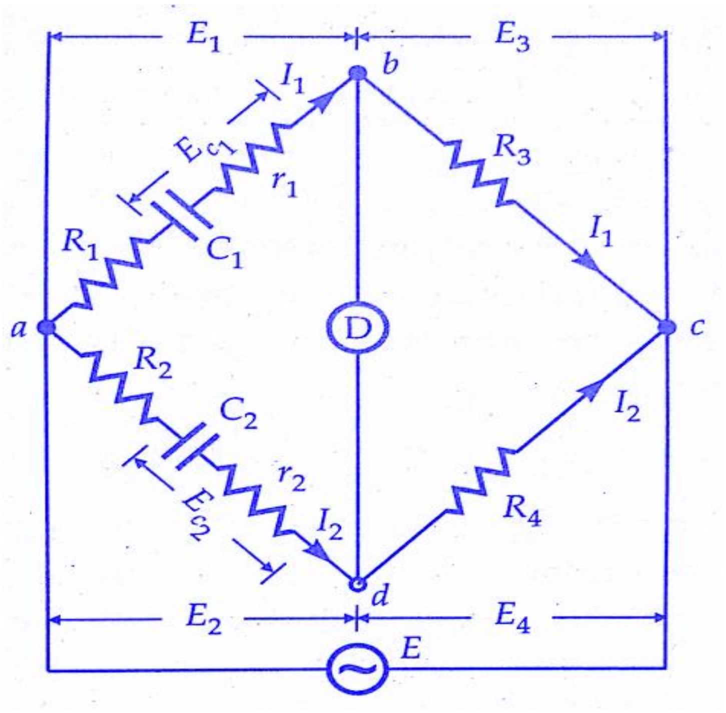


Phasor Diagram of De-Sauty’s Bridge

- In order to make measurement of imperfect capacitor (Capacitors having dielectric loss), the bridge is modified.
- Resistors R_1 and R_2 are connected in series with C_1 and C_2 respectively. The r_1 and r_2 are resistances representing the loss component of the two capacitors.
- At balance,

$$\left(R_1 + r_1 + \frac{1}{j\omega C_1}\right) R_4 = \left(R_2 + r_2 + \frac{1}{j\omega C_2}\right) R_3 \quad (5)$$

- After simplification of eq. (5)

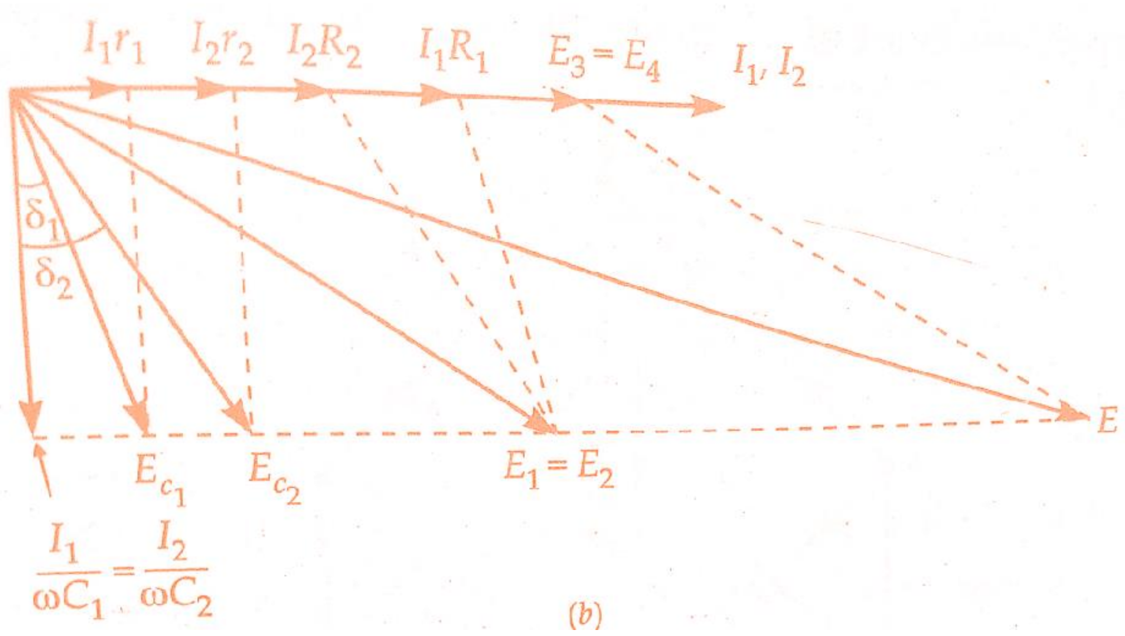


Modified De Sauty's Bridge

$$\frac{C_1}{C_2} = \frac{R_2 + r_2}{R_1 + r_1} = \frac{R_4}{R_3} \quad (6)$$

- The balance may be obtained by variation of resistance $R_{1,}$ R_2, R_3 and R_4.

- The phasor diagram of modified De Sauty's bridge.

- 

- The angles δ_1 and δ_2 are the phase angles of **capacitors** C_1 and C_2 respectively.
- Dissipation factors for the capacitors are :

$D_1 = \tan \delta_1 = \omega C_1 r_1$ and $D_2 = \tan \delta_2 = \omega C_2 r_2$

- From eq. (6)

$$\frac{C_1}{C_2} = \frac{R_2 + r_2}{R_1 + r_1}$$

or $$C_2 r_2 - C_1 r_1 = C_1 R_1 - C_2 R_2$$

or $$\omega C_2 r_2 - \omega C_1 r_1 = \omega(C_1 R_1 - C_2 R_2)$$

$\therefore$ $$D_2 - D_1 = \omega(C_1 R_1 - C_2 R_2)$$

But $$C_1 / C_2 = R_4 / R_3$$

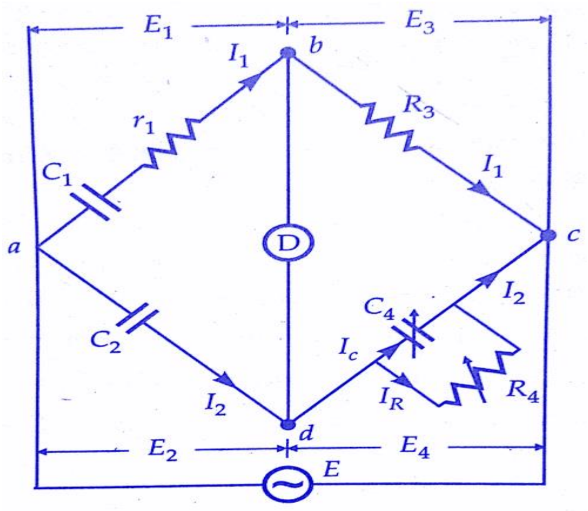
$\therefore$ $$C_1 = C_2 R_4 / R_3$$

Hence $$D_2 - D_1 = \omega C_2 \left(\frac{R_1 R_4}{R_3} - R_2 \right)$$

- If the **dissipation factor** of one of the capacitors is known, the dissipation factor for the other can be determined.
- **De Sauty's Bridge** does not give accurate results for dissipation factor since its value depends on the difference of quantities R_1R_4/R_3 and R_2.

Schering Bridge

- Schering Bridge is used to measure unknown capacitance by comparing with standard capacitance.
- C_1 = Capacitor whose **capacitance is to be determined.**
- r_1 = A series resistance representing the loss in the capacitor.
- C_2 = A standard capacitor. This capacitor is either an air or a gas capacitor and hence is loss-free.
- R_3 = A non-inductive resistance.
- C_4 = A standard variable capacitor.
- R_4 = A variable non-inductive resistance in parallel with variable capacitor C_4.



- At balanced condition-

$$Z_1 Z_4 = Z_2 Z_3 \quad (1)$$

Impedance of arm ‘**ab**’ $Z_1 = \left(r_1 + \frac{1}{j\omega C_1}\right)$

Impedance of arm ‘**ad**’ $Z_2 = 1/j\omega C_2$

Impedance of arm ‘**bc**’ $Z_3 = R_3$

Impedance of arm ‘**cd**’ $Z_4 = \left(\frac{R_4}{1 + j\omega R_4 C_4}\right)$

(2)

- Put the expression of eq. (2) into eq. (1)

$$\left(r_1 + \frac{1}{j\omega C_1}\right)\left(\frac{R_4}{1 + j\omega R_4 C_4}\right) = \frac{1}{j\omega C_2} R_3 \qquad (3)$$

$$\left(r_1 + \frac{1}{j\omega C_1}\right) R_4 = \frac{R_3}{j\omega C_2}(1 + j\omega C_4 R_4)$$

$$r_1 R_4 - \frac{jR_4}{\omega C_1} = -j\frac{R_3}{\omega C_2} + \frac{R_3 R_4 C_4}{C_2} \qquad (4)$$

- Equating the real and imaginary terms-

$r_1 = R_3 C_4 / C_2$

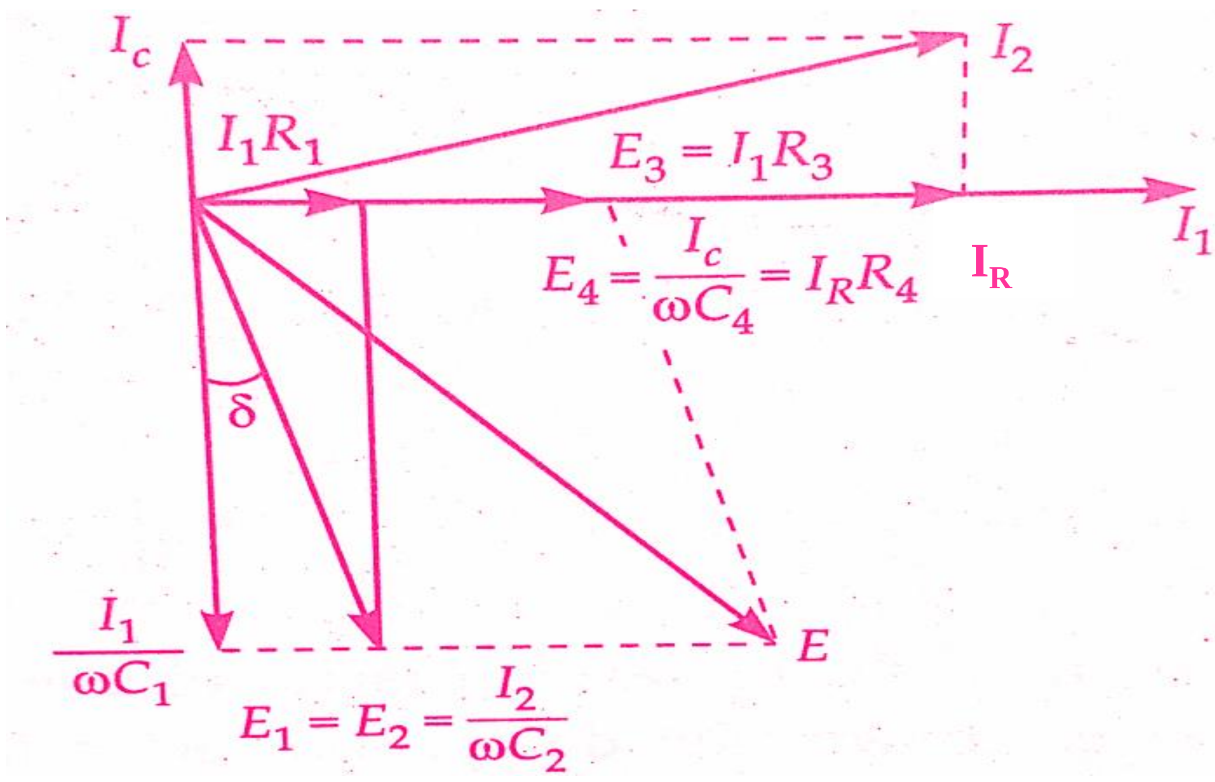
and $C_1 = C_2 \, (R_4 / R_3)$

- Two independent balance equations are obtained if C_4 and R_4 are chosen as the variable elements.

- **Dissipation factor,**

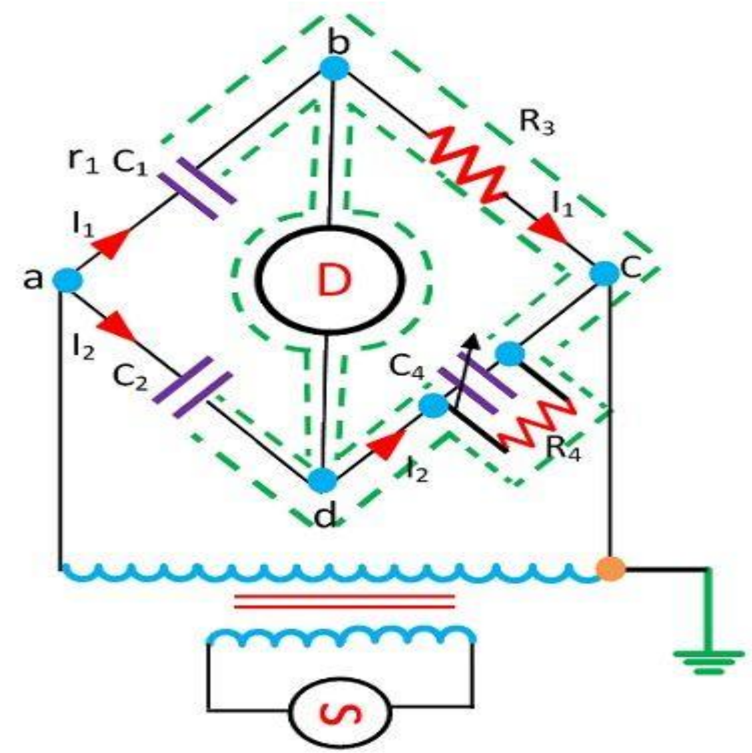
  $D_1 = \tan\delta = \omega C_1 r_1 = \omega (C_2R_4/R_3) \times (R_3C_4/C_2) = \omega C_4R_4$

- The values of capacitance C_1 and its dissipation factor are obtained from the values of bridge elements at balance.
- Phasor Diagram of Schering bridge-



High Voltage Schering Bridge

- The high voltage Schering bridge is used to measure the unknown capacitance, relative permeability, dissipation factor, dielectric loss of a capacitor, measurement of the properties of insulators, capacitor bushings, insulating oil etc.
- This bridge is suitable for measuring the small capacitance supplied by high frequency and high voltage source.



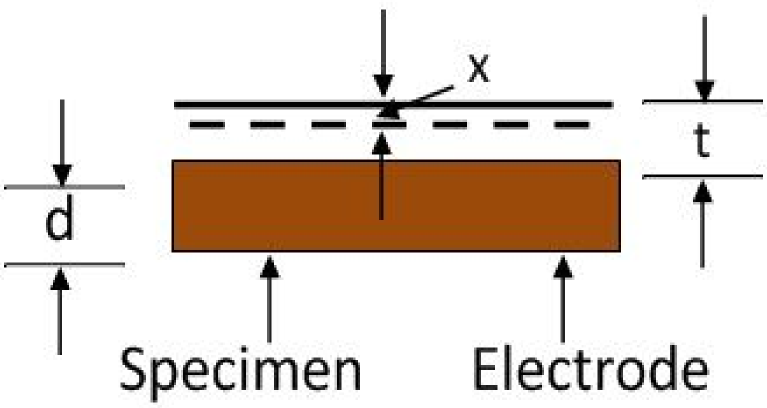
- High voltage supply is obtained from a transformer and the detector is a vibration galvanometer.

- The high voltage working capacitors are placed in the arms **ab** and **ad.** The impedance of the arm **ab** and **ad** are very high as compared to the arm **bc and cd.** The point c is earthed.
- Major portion of potential drop will be in the **ab** and **ad** arms and very little voltage drop occurs in **bc** and **dc** arm.
- Little voltage drop across **bc** and **dc** is advantageous because controls are located at these branch and for the safety of the operator.
- The potential across the detector is also kept low.
- The spark gap sets are placed on each arm of **bc** and **dc** for preventing the dangerous due to breakdown of the high voltage capacitors.
- The power loss is very small in the arms **ab** and **ad** because of the high impedance (less current will be drawn) of arms **ab and ad.**

Measurement of Relative Permittivity with Schering Bridge

- Relative Permittivity is determined by measurement of capacitance of small capacitor with the specimen as dielectric.
- Parallel plate or concentric cylinder forms the electrodes of the capacitor.
- In normal arrangement, solid material used as disc specimen with metal electrodes.
- The electrodes may consists of
  - Thin metal foil attached to specimen by petroleum jelly.
  - Thin films of silver or aluminium applied by evaporation.
  - Mercury electrodes are obtained by floating the specimen on the surface of a mercury pool.
- The relative permittivity is calculated from the measured value of capacitance and the dimensions of the electrodes.
- For a parallel arrangement, relative permittivity is expressed as:
- $\varepsilon_r = \frac{C_s d}{\varepsilon_0 A}$ d – Spacing between electrodes
- A – Effective area of electrodes. ε_0 – permittivity of free space
- C_s – measured value of capacitance by considering the specimen as a dielectric.

- To avoid the close contact to electrodes and specimen, guard ring is inserted between them.
- First capacitance is measured and than remove the specimen and spacing between the electrodes is adjusted until the capacitance is the same as before.
- The relative permittivity of the specimen can be calculated from the thickness of the specimen and the change in electrode spacing.



Let

C – Capacitance between the electrode and specimen
A – Area of electrodes
d – Thickness of the specimen
t – Gap between specimen and electrode.
x – Reduction in separation between the specimen and electrode.
C_s – Capacitance of specimen.
C_0 – Capacitance between the space because of specimen and electrode.
C – Effective capacitance of C_s and C_0.

- The capacitance between the specimen and the electrode is expressed as

$$C = \frac{C_s C_0}{C_s + C_0} = \frac{\left(\frac{\varepsilon_r \varepsilon_0 A}{d}\right) \cdot \left(\frac{\varepsilon_0 A}{t}\right)}{\left(\frac{\varepsilon_r \varepsilon_0 A}{d}\right) + \left(\frac{\varepsilon_0 A}{t}\right)} = \frac{\varepsilon_r \varepsilon_0 A}{\varepsilon_r t + d}$$

- When we reduce the specimen and the spacing is again adjusted for obtaining the same value of capacitance, the expression for reducing specimen is-

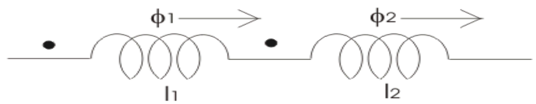
$$C = \frac{\varepsilon_r \varepsilon_0 A}{\varepsilon_r t + d}$$

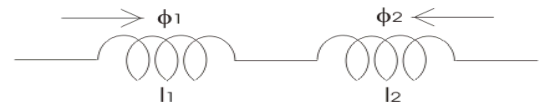
$$\frac{\varepsilon_o\, A}{t + d - x} = \frac{\varepsilon_r \varepsilon_o}{\varepsilon_r t + d}$$

$$\varepsilon_r = \frac{d}{d - x}$$

Heaviside Bridge

- Two coils are connected in series as-





- Such that the magnetic fields are additive, the resultant inductor of these two can be calculated as

$$L_x = L_1 + L_2 + 2M \qquad (1)$$

  - Where, L_1 is the self inductor of first coil, L_2 is the self inductor of second coil, M is the mutual inductor of these two coils.

- Now if the connections of any one of the coils is reversed then-

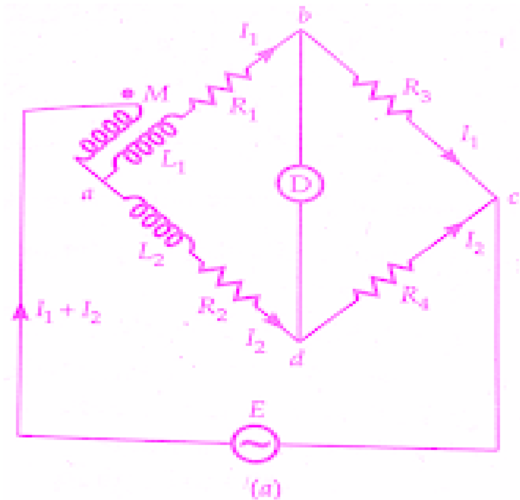
$$L_y = L_1 + L_2 - 2M \qquad (2)$$

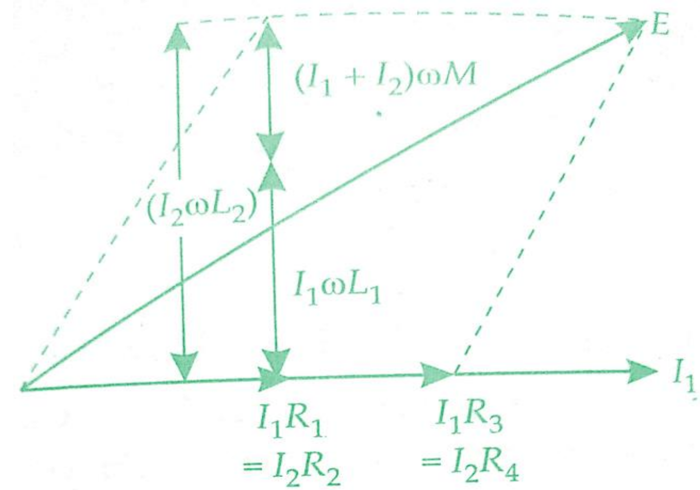
- On solving these two equations-

$$M = \frac{L_x - L_y}{4}$$

- The mutual inductor of the two coils connected in series is given by one-fourth of the difference between the measured value of self inductor when taking the direction of field in the same direction and value of self inductor when the direction of field is reversed.

- **Heaviside Bridge measures mutual inductance** in terms of a known self-inductance.





  - M = unknown mutual inductance,
  - L_1= self-inductance of secondary coil whose **mutual inductance** has to be measured.
  - L_2 = known self-inductance.
  - R_1, R_2, R_3 and R_4 = non-inductive resistors.
- At balance, voltage drop between b and c must equal the voltage drop between d and c.
- Also, the voltage drop across a-b-c must equal the voltage drop across a-d-c.

- Thus following equations are obtained at balance-

$$I_1 R_3 = I_2 R_4$$

$$(I_1 + I_2)(j\omega M) + I_1(R_1 + R_3 + j\omega L_1)$$

$$= I_2(R_2 + R_4 + j\omega L_2)$$

$$\therefore \quad I_2\left(\frac{R_4}{R_3} + 1\right)j\omega M + I_2 \frac{R_4}{R_3}(R_1 + R_3 + j\omega L_1)$$

$$= I_2(R_2 + R_4 + j\omega L_2)$$

$$j\omega M\left(\frac{R_4}{R_3} + 1\right) + \frac{R_4}{R_3} R_1 + R_4 + j\omega L_1 \frac{R_4}{R_3} = R_2 + R_4 + j\omega L_2$$

Separating out real and imaginary terms-

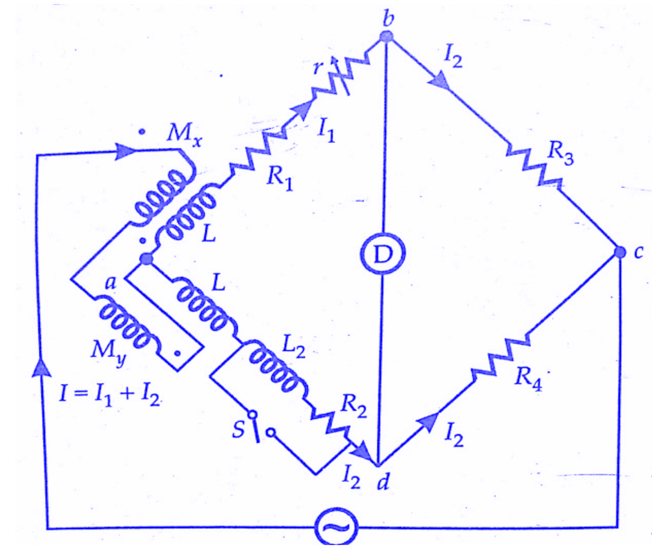
$$R_1 = R_2 R_3 / R_4$$

$$M = \frac{L_2 - L_1 R_4 / R_3}{R_4 / R_3 + 1} = \frac{R_3 L_2 - R_4 L_1}{R_3 + R_4}$$

Campbell's Modification of Heaviside Bridge

- This is used to *measure a self-inductance* in terms of a **mutual inductance**.
- In this case, an additional balancing coil R is included in arm a-d in series with inductor under test.
- An additional resistance r is put in arm a-b.
- Balance is obtained by varying M and r.

  A short circuiting switch is placed across the coil R2, L2 under measurement.
- Two sets of readings are taken one with switch being open and the other with switch being closed.Let values of M and r, be M1 and r1 with switch open, and M2 and r2 with switch closed.



- With switch open

$$L_2 + L = \frac{M_1(R_3 + R_4) + R_4 L_1}{R_3}$$

- With switch close

$$L = \frac{M_2(R_3 + R_4) + R_4 L_1}{R_3}$$

$$L_2 = (M_1 - M_2)(1 + R_4/R_3)$$

- With switch open
- $$R_2 + R = (R_1 + r_1)R_4/R_3$$

- and with the switch closed

$$R = (R_1 + r_2)R_4/R_3$$

$$R_2 = (r_1 - r_2)\, R_4/R_3$$